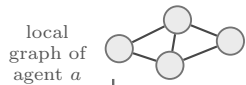


1. Encode



Node projection
Linear $d_v \rightarrow d_h$, ReLU, LN

$K \times$ message-passing layers
configurable graph encoder
 $\mathbf{H}^{(0)} \rightarrow \mathbf{H}^{(K)}$

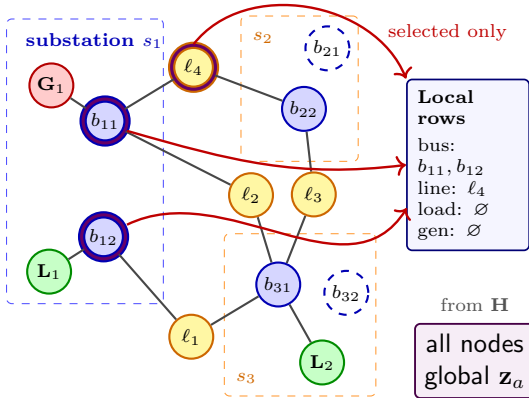
$\mathbf{H}$: one state per node

Selected-node readout + projection
 $\Rightarrow$ global $\mathbf{z}_a$

shared encoder: one weight set for every agent

2. Gather the touched rows

candidate j : move the origin of line ℓ_4 from b_{11} to b_{12}



3. Choose one action context

Joint mean

One mean over all touched node states;
output width d_h .

Type-separated mean

Mean by type, then concatenate
[bus; line; $\mathbf{0}_{\text{load}}$; $\mathbf{0}_{\text{gen}}$]; width $4d_h$.

4. Score the candidate

$\mathbf{z}_a$
global

ϕ_j
optional

$\mathbf{c}_j$

$[\mathbf{z}_a; \mathbf{c}_j; (\phi_j)]$

$\text{MLP}_{\text{score}}$ or $\text{MLP}_{\text{score}}^{(a)}$
shared across agents or agent-specific

$\ell_{a,j}$

one logit, $j \geq 1$

Idle action $j = 0$: choose one branch

Shared: $\mathbf{c}_0 = \mathbf{0}$ or
Dedicated: $\mathbf{z}_a$ only

$+b_\emptyset$
 $\ell_{a,0}$



touched by candidate j

— graph relation



optional input